Dehazing-DiffGAN: Sequential Fusion of Diffusion Models and GANs for High-Fidelity Remote Sensing Image Dehazing

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***Abstract*—Atmospheric haze significantly reduces image quality by obscuring important visual details, which in turn impairs the performance of computer vision tasks. Many existing dehazing methods struggle in heavy haze conditions, often resulting in images with poor visual quality, unrealistic features, or visible artifacts. To overcome these limitations, we propose Dehazing-DiffGAN, a novel deep learning framework that combines diffusion models with generative adversarial networks (GANs) in a dual-phase pipeline. In the dehazing phase, the diffusion model progressively removes haze by simulating atmospheric scattering, effectively capturing both local structures and global context. In the enhancement phase, a GAN-based enhancement module improves image quality by emphasizing fine details and textures, enabling precise restoration of edges and high-frequency components. This synergistic fusion approach allows Dehazing-DiffGAN to generate images of superior visual quality, often indistinguishable from real, haze-free images. Extensive experiments show that Dehazing-DiffGAN achieves state-of-the-art performance on several challenging benchmarks. On the high-resolution SateHaze1k dataset for remote sensing, our model achieves a peak signal-to-noise ratio (PSNR) of 27.21 dB, outperforming the next best method at 26.84 dB, and obtains a superior learned perceptual image patch similarity (LPIPS) score of 0.0472 compared to 0.08. On the ultrahigh-resolution DNH-HAZE dataset (6000 × 4000 pixels), Dehazing-DiffGAN achieves the best LPIPS score of 0.267, significantly improving over the second-best score of 0.326. These results highlight the effectiveness of Dehazing-DiffGAN in improving image clarity and confirm its robustness and adaptability across a variety of dehazing tasks. The code is available at https://github.com/riotulab/Dehazing-DiffGAN**

***Index Terms*—Deep learning, dual-domain processing, encoder–decoder architecture, high-resolution images, image dehazing, image enhancement, nonhomogeneous haze, remote sensing.**

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I. Introduction

IMAGE dehazing, particularly in nonhomogeneous scenes, represents a significant challenge in computer vision. Atmospheric haze degrades image quality by obscuring critical details and impairing the performance of vision tasks such as classification, object detection, tracking, and semantic segmentation, thereby reducing the accuracy and reliability of data interpretation [1]. Moreover, distinguishing between haze and natural sky color becomes particularly difficult in the presence of nonhomogeneous haze. The inherent complexity of haze, which scatters light and applies blur nonuniformly across a scene, calls for models capable of authentically reconstructing visual information for further processing [2], [3].

In practice, the attenuation of light due to haze is encapsulated by a standard physical model that describes how light undergoes absorption, scattering, and attenuation when interacting with haze particles. This model is mathematically depicted as follows:

$$I_h(\mathbf{x}) = t(\mathbf{x})\, I_0(\mathbf{x}) + (1 - t(\mathbf{x}))\, L \tag{1}$$

where $I_h(\mathbf{x})$ represents the hazy image captured by the sensor, $I_0(\mathbf{x})$ represents the pristine image before haze interference, $t(\mathbf{x})$ represents the transmission function showing the haze's impact, and L represents the ambient atmospheric illumination. The functional form of the transmission function is

$$t(\mathbf{x}) = e^{-\beta d(\mathbf{x})} \tag{2}$$

where β denotes the atmospheric scattering coefficient, and $d(\mathbf{x})$ indicates the scene depth, varying with pixel location [4], [5]. Early studies in dehazing primarily employed simplistic models that estimated atmospheric scattering through basic or complex prior-based methods, such as the dark-channel prior and more intricate nonlocal approaches. These models, however, typically assume a uniform distribution of haze, overlooking the real-world complexity and spatial variability of haze dispersion.

Despite the significant progress driven by deep learning techniques in the field of image dehazing [6], several substantial challenges remain unresolved, particularly in the effective integration of global and local features while maintaining high perceptual quality in haze-free images. Traditional dehazing models often struggle to accurately capture and harmonize the intricate interactions between global and local features,

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a capability that is critical for producing high-quality remote sensing images [7]. This limitation is particularly evident in applications where the fidelity of fine details is crucial for subsequent analysis and decision-making.

Over recent years, deep learning-based approaches have made notable strides in image restoration, including dehazing, attempting to recover high-quality imagery from compromised inputs [8], [9], [10], [11], [12]. However, these methods frequently rely on pixel-based reconstruction loss functions, such as the L_1/L_2 norms, resulting in over-smoothed images that often lack fine details, thereby diminishing the perceptual quality of the restored images. Advancements in GANs [13], [14], [15], [16], which exhibit a capacity for producing images of strikingly high quality. Nonetheless, these GAN-based models can introduce geometric distortions, particularly around sharp edges and intricate textures. Methods that emphasize peak signal-to-noise ratio (PSNR) may yield images with correctly aligned but blurred edges, while those prioritizing perceptual quality can produce images with sharper, yet potentially misaligned edges. The unpredictability brought about by the discriminator components in these networks is a significant contributor to structural inconsistencies. Although attempts like the controllable feature space network (CFSNet) [17] aim to balance perceptual quality and distortion minimization, achieving simultaneous optimization for detailed texture fidelity and structural integrity remains challenging.

Furthermore, the increasing complexity of modern neural network architectures, such as vision transformers (ViTs), presents additional challenges. While these architectures have demonstrated remarkable potential in various computer vision tasks [18], [19], they often encounter difficulties related to computational efficiency and the effective synthesis of features across different scales and regions of the image. The high computational demands and intricate design of these models can hinder their practical application, especially in scenarios requiring real-time processing or deployment on resource-constrained platforms.

Although recent research has introduced models that attempt to internally combine global and local features, such as those employing advanced attention mechanisms [20], [21], these approaches still encounter significant limitations, particularly when addressing nonhomogeneous haze. Nonhomogeneous haze presents a highly complex challenge due to its spatially varying density and its visual similarity to natural sky regions. Many existing models struggle to accurately differentiate between haze and the inherent sky color, leading to incomplete haze removal where residual artifacts persist in certain parts of the image. Developing an effective framework remains difficult, as it requires not only the seamless integration of global and local features but also the precise preservation and enhancement of visual fidelity under complex haze conditions. This challenge is especially critical in remote sensing applications, where the accurate reconstruction of fine details and atmospheric consistency is paramount for reliable interpretation and analysis.

In response to these challenges, we present Dehazing-DiffGAN, a novel dual-phase deep learning architecture developed to improve the efficiency of image dehazing and enhancement. The proposed model demonstrates superior performance across various datasets. The model comprises a dehazing phase utilizing an encoder–decoder structure with novel normalized attention fusion blocks (NAFBlocks), followed by a dual-domain enhancement phase. In the dehazing phase, an encoder–decoder network with three stages of downsampling and upsampling is employed, interconnected by NAFBlocks. These blocks integrate layer normalization, convolutions, SimpleGate mechanisms, and spatial context-aware modules to effectively capture both local and global contextual information. The enhancement phase introduces a dual-domain approach where the dehazed image is processed in parallel in the RGB domain and the Laplacian domain. The Laplacian domain emphasizes edge information, which is crucial for enhancing structural details. Each branch begins with a 3×3 convolution followed by multiple residual-in-residual dense blocks (RRDBs) for deep feature extraction. The enhanced features from both domains are then fused through a fusion block to produce the output image. This dual-domain approach facilitates the simultaneous improvement of low-frequency content and high-frequency edge features, resulting in enhanced image quality. The Dehazing-DiffGAN demonstrates superior perceptual quality on two challenging datasets: the nonhomogeneous dehazing dataset from NTIRE and the remote sensing-focused SateHaze1k dataset. These results underscore the model's robustness and generalizability across varied dehazing scenarios, highlighting its potential for both general and specialized remote sensing applications.

The main contributions of this article are summarized as follows.

1) We propose Dehazing-DiffGAN, a novel fusion architecture combining diffusion models and GANs for efficient image dehazing and enhancement.
2) The dehazing phase uses an encoder–decoder structure with NAFBlocks to capture local and global contextual information, effectively addressing heavy haze.
3) The enhancement phase processes the dehazed image concurrently by fusing the content from the RGB and the Laplacian domains, enhancing both the learning of low-frequency content and high-frequency edge details for superior image quality.
4) Our method achieves state-of-the-art performance on challenging datasets, such as the NTIRE dehazing dataset and SateHaze1k, surpassing previous models in terms of PSNR and learned perceptual image patch similarity (LPIPS) scores.
5) Comprehensive experiments validate Dehazing-DiffGAN's effectiveness, demonstrating its robustness and potential for various dehazing applications.

By effectively addressing the complexities of atmospheric haze in remote sensing imagery, *Dehazing-DiffGAN* enhances the utility of RS data for critical applications requiring high precision and detail. The remainder of this article is organized as follows. Section II reviews related work in image dehazing and attention mechanisms. Section III presents the proposed methodology, detailing the architecture of *Dehazing-DiffGAN* and its components. Section IV discusses the results and provides a comprehensive discussion, including comparisons

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with state-of-the-art methods. Finally, Section V concludes this article and suggests directions for future research.

II. Related Work

The field of image dehazing has witnessed significant advancements with the advent of deep learning technologies. Early models based on convolutional neural networks (CNNs), such as SRCNN [8] and DnCNN [9], demonstrated the potential of CNNs in image restoration tasks, including dehazing. These architectures often used pixel reconstruction losses such as norms L_1 and adversarial losses from GANs to refine image quality and detail [22], [23], [24], [25]. However, due to their inherently local receptive fields, these early models were limited in capturing long-range dependencies and global contextual information, which are crucial for handling the complex scattering effects present in hazy images.

To address the limitations of CNNs in modeling global context, transformer architectures were integrated into image restoration research, marking a significant shift in the field. Transformers, recognized for their effectiveness in high-level vision tasks [18], were adapted for image processing applications. Models such as IPT [26], SwinIR [27], SGLC [28], Restormer [29], and Uformer [30] introduced innovative strategies by merging transformer capabilities with CNNs. These hybrid models leveraged the global context modeling capabilities of transformers alongside the local feature extraction capabilities of CNNs, offering novel solutions to complex challenges such as image dehazing. However, transformer-based models often come with increased computational complexity and memory requirements, making them less suitable for high-resolution images or real-time applications.

To address the challenges of high-resolution image dehazing, several specialized models have been developed to capture the fine-grained semantic and structural details inherent in high-resolution images. DHGAN [31] aimed to extract global features effectively, while BEGAN [32] focused on enlarging the receptive field to capture more context. High-resolution dehazing models like HR-Dehazer [33] and 4K-Dehazer [7] tailored their architectures accordingly. H2RL-Net [34] adopted a dual-branch approach, incorporating multiresolution CNNs and advanced feature refinement to preserve details and semantic integrity. However, these models often struggled to balance computational efficiency and dehazing performance, especially when dealing with extremely high-resolution images, which posed challenges for practical deployment in resource-constrained environments.

Moreover, diffusion models emerged as a novel approach in image restoration [35], [36], offering iterative refinement of images through learned noise models. This method showed promise in enhancing the perceptual quality of dehazed images by effectively modeling complex distributions. Nevertheless, diffusion models typically require numerous iterative steps to achieve high-quality results, leading to increased computational demands and processing times that may not be feasible for real-time applications.

In the realm of remote sensing and nonhomogeneous image dehazing, where atmospheric conditions and complex scenes present unique challenges, several approaches have been proposed. Qin et al. [37] introduced FFA-Net, employing a Feature Fusion Attention mechanism to effectively integrate features from different network layers, resulting in clearer and more detailed dehazed images. Guo et al. [38] developed SCANet, a self-paced semi-curricular attention network specifically designed for nonhomogeneous image dehazing. SCANet introduced a novel training strategy in which the network progressively focuses on more challenging tasks, effectively handling varying degrees of haze within a single image. However, these models may not fully capture the complex atmospheric scattering effects in remote sensing images, and their performance can degrade when faced with highly variable haze conditions.

To further enhance detail preservation, Dong et al. [39] proposed a detail-enhanced dehazing network utilizing multiscale detail refinement units and hierarchical attention mechanisms. This approach significantly improved the network's capability to recover intricate details and textures in dehazed images, enhancing both clarity and perceptual quality. However, while this method enhances details, it may not adequately address the balance between detail preservation and global contrast, sometimes leading to unnatural appearances in the dehazed images.

In addition, Huang et al. [40] advanced dehazing precision in remote sensing images using conditional generative adversarial networks (cGANs), particularly excelling in challenging conditions where traditional methods often fall short. However, cGAN-based approaches can suffer from training instability and may introduce artifacts into the restored images. Li et al. [41] introduced GTMNet, a hybrid model that combines various dehazing strategies to enhance the effectiveness and reliability of dehazing processes in remote sensing imagery. Although effective, hybrid models such as GTMNet involve complex architectures that are computationally demanding, potentially limiting their applicability in real-time or large-scale processing scenarios.

Recent studies have also revisited the capabilities of CNN architectures. Cui et al. [42] introduced the ConvIR model, demonstrating that optimized convolutional architectures can compete with, and sometimes surpass, transformer-based models in tasks such as dehazing and deblurring. Their extensive testing across multiple benchmark datasets and restoration scenarios underscored the continued relevance and efficiency of CNNs. Despite these promising results, CNN-based models may still face challenges in capturing long-range dependencies and global context as effectively as transformer-based models, potentially limiting their performance on complex scenes with intricate haze distributions.

Lihe et al. [43] presented PhDnet, which combines residual learning with atmospheric scattering models to improve interpretability and effectiveness, achieving superior results under various atmospheric conditions. However, integrating physical models can increase the complexity of the network and may require precise atmospheric parameters, which are frequently challenging to acquire in practical situations.

Moreover, Yu et al. [44] developed HSMD-Net, leveraging intralayer and interlayer feature correlations to improve

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color and texture accuracy while preserving semantic integrity. Huang et al. [45] proposed the adaptive diffusion null-space dehazing network (ADND-Net), utilizing a diffusion-model-based null-space learning strategy to manage uneven haze in large remote sensing images, ensuring enhanced consistency and quality. Nevertheless, these methods can be computationally intensive due to their complex feature correlation computations and diffusion processes, posing challenges for real-time or resource-constrained applications.

Despite these advancements, challenges remain in effectively integrating global and local features while maintaining high perceptual quality in dehazed images, especially in high-resolution and remote sensing applications. Existing models often struggle to harmonize the intricate interactions between global context and local details, which is critical to producing high-quality dehazed images with fine structural attributes necessary for subsequent analysis and decision-making.

To address these challenges, we propose the **Dehazing-DiffGAN**, an innovative dual-phase deep learning architecture designed for efficient image dehazing and enhancement. The Dehazing-DiffGAN comprises a dehazing phase utilizing an encoder–decoder structure with novel NAFBlocks, followed by a dual-domain enhancement phase. In the dehazing phase, the network effectively captures both local and global contextual information. The enhancement phase processes the dehazed image concurrently in the RGB domain and the Laplacian domain, allowing simultaneous enhancement of low-frequency content and high-frequency edge details. This approach yields superior image quality, demonstrating state-of-the-art performance across diverse datasets, and underscores the model's robustness and generalizability in various dehazing scenarios.

III. Proposed Methodology

In addressing the challenge of dense and nonhomogeneous haze removal, our proposed framework, Dehazing-DiffGAN, is designed with two complementary stages that work together to enhance image restoration and quality, as shown in Fig. 1. The methodology integrates a diffusion-based dehazing phase followed by a GAN-based enhancement phase, leveraging the unique strengths of each model to achieve superior results.

The dehazing phase employs a diffusion model that progressively refines the image by simulating the natural process of diffusion. This approach systematically removes haze by capturing both broad scene-level (global) patterns and fine-grained (local) textures, effectively restoring the underlying image structure. This can be likened to the gradual clearing of fog from a window, revealing both the overall landscape and finer details. However, due to the complex nature of atmospheric modeling, this phase can sometimes introduce minor distortions or noise.

To mitigate these artifacts, the enhancement phase utilizes a GAN, which further refines the image by focusing on enhancing textures and edges. This phase acts like a polishing process, bringing out sharper details and ensuring a visually appealing, high-quality result. By combining the strengths of both the diffusion model and GAN, Dehazing-DiffGAN achieves a balanced integration of large-scale atmospheric correction with the preservation of fine details, making it particularly effective for high-resolution remote sensing applications. Sections III-A and III-B delve into the technical implementation of each stage.

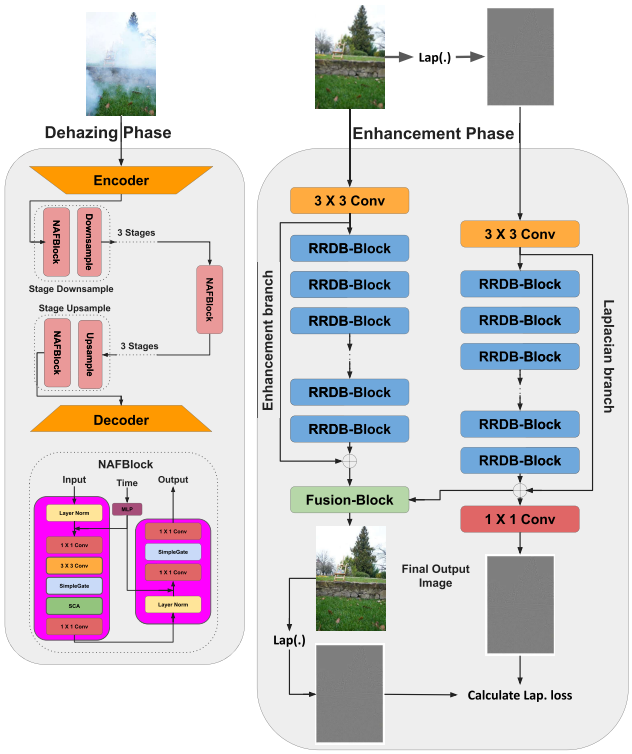


Fig. 1. Dehazing-DiffGAN schematic.

A. Dehazing Phase

The initial phase of our proposed Dehazing-DiffGAN leverages a diffusion-based model to restore hazy images to their high-quality counterparts. This approach is grounded in stochastic differential equations (SDEs) and incorporates a modified nonlinear activation free network (NAFNet) within a latent diffusion framework to efficiently handle high-resolution images.

We model the degradation and restoration processes using an SDE with mean-reversion properties. The forward diffusion process replicates the degradation of a high-quality image into a hazy version and is defined as

$$\mathrm{d}\mathbf{x} = \theta_t\,(\boldsymbol{\mu} - \mathbf{x})\,\mathrm{d}t \;+\; \sigma_t\,\mathrm{d}\mathbf{w} \tag{3}$$

where $\mathbf{x}$ indicates the image state at time t, $\boldsymbol{\mu}$ represents the equilibrium condition corresponding to the hazy (degraded) state, the high-quality image, θ_t denotes the rate of mean-reversion that varies with time, σ_t is the volatility coefficient dependent on time, and $\mathrm{d}\mathbf{w}$ refers to the standard Wiener process. This equation models how an image degrades over time, converging toward the equilibrium state $\boldsymbol{\mu}$. By setting $(\sigma_t^2/\theta_t) = 2\lambda^2$, with λ being a constant, we can analytically derive the marginal distribution $p_t(\mathbf{x})$ [46]

$$p_t\,(\mathbf{x}) = \mathcal{N}\,(\mathbf{x}; \boldsymbol{m}_t, v_t\mathbf{I}) \tag{4}$$

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TABLE I

FEATURE-MAP DIMENSIONS AND DEPTH OF EACH BLOCK IN THE LATENT-SPACE ENCODER–DECODER USED IN THE DEHAZING PHASE. R DENOTES THE DOWN-SAMPLING RATIO WITH RESPECT TO THE INPUT RESOLUTION $H \times W$

Stage l	**Spatial size** $(H_l \times W_l)$	**Channels** C_l	**NAFBlocks** k_l
Enc-0	$H \times W$	64	2
Enc-1	$\frac{H}{2} \times \frac{W}{2}$ $(R{=}2)$	128	2
Enc-2	$\frac{H}{4} \times \frac{W}{4}$ $(R{=}4)$	256	4
Bottleneck	$\frac{H}{8} \times \frac{W}{8}$ $(R{=}8)$	512	6
Dec-2	$\frac{H}{4} \times \frac{W}{4}$	256	4
Dec-1	$\frac{H}{2} \times \frac{W}{2}$	128	2
Dec-0 / Out	$H \times W$	64	2

where

$$\boldsymbol{m}_t = \boldsymbol{\mu} + (\mathbf{x}_0 - \boldsymbol{\mu})\, e^{-\bar{\theta}_t} \tag{5}$$

$$v_t = \lambda^2 \left(1 - e^{-2\bar{\theta}_t}\right) \tag{6}$$

$$\bar{\theta}_t = \int_0^t \theta_s \, \mathrm{d}s \tag{7}$$

and $\mathbf{x}_0$ is the initial state at $t = 0$. As time progresses, $\mathbf{x}$ converges to $\boldsymbol{\mu}$ with variance λ^2, effectively modeling the degradation process from a high-quality image to a hazy one.

To restore the image, we consider the reverse-time SDE, which inverts the degradation process

$$\mathrm{d}\mathbf{x} = \left[\theta_t (\boldsymbol{\mu} - \mathbf{x}) - \sigma_t^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x})\right] \mathrm{d}t + \sigma_t \mathrm{d}\bar{\mathbf{w}} \tag{8}$$

where $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ is the score function, representing the gradient of the log-probability density with respect to $\mathbf{x}$, and $\mathrm{d}\bar{\mathbf{w}}$ is the reverse-time Wiener process [47]. The score function guides the restoration by indicating the direction in which the image needs to be adjusted at each time step to recover the high-quality image.

However, directly applying diffusion models to high-resolution images is computationally intensive due to the large size of the data. To address this challenge, we perform the diffusion process in a lower dimensional latent space using an encoder–decoder architecture [35]. Specifically, we encode the high-resolution hazy image $\mathbf{I}_{\text{hazy}}$ into a latent representation $\mathbf{z}_0 = E(\mathbf{I}_{\text{hazy}})$ using an encoder $E(\cdot)$. The diffusion process is then applied within the latent space

$$\mathrm{d}\mathbf{z} = \theta_t \left(\boldsymbol{\mu}_{\mathbf{z}} - \mathbf{z}\right) \mathrm{d}t + \sigma_t \, \mathrm{d}\mathbf{w}_{\mathbf{z}} \tag{9}$$

where $\mathbf{z}$ is the latent code, and $\boldsymbol{\mu}_{\mathbf{z}}$ is the equilibrium latent state. By operating in the latent space, we significantly reduce computational requirements while retaining essential information for accurate dehazing. The restored latent code $\mathbf{z}_{\text{restored}}$ is then decoded to reconstruct the high-quality image

$$\mathbf{I}_{\text{restored}} = D\left(\mathbf{z}_{\text{restored}}\right). \tag{10}$$

Using a decoder $D(\cdot)$, skip connections linking the encoder and decoder layers facilitate the transfer of feature maps across corresponding resolutions, maintaining multiscale details and spatial information. This mechanism is crucial for restoring fine textures and structural details in dehazed images.

Table I enumerates, for every stage l of the latent U-Net: 1) the spatial resolution $(H_l \times W_l)$; 2) the channel width C_l; and 3) the number of stacked *NAFBlocks* k_l. Down-sampling is realized by a 3×3 convolution with stride 2, whereas up-sampling consists of nearest-neighbor resize followed by a 3×3 convolution.

Skip connections copy the feature tensor $\mathbf{F}^{(l)} \in \mathbb{R}^{C_l \times H_l \times W_l}$ from encoder to decoder at the same resolution, so the decoder receives the concatenation $\left[\text{upsampled output} \parallel \mathbf{F}^{(l)}\right]$ and can fuse global context with the high-frequency detail preserved by the encoder.

Empirically, this layout gives the best tradeoff between perceptual quality and memory footprint on 6000 × 4000 DNH-HAZE images (where the network processes H = 512, W = 512 crops) and on full-resolution 512 × 512 SateHaze1k scenes.

A critical component of the diffusion model is the noise estimation module, which is responsible for approximating the score function by predicting the noise added at each timestep. To ensure both computational efficiency and accurate restoration, we adopt the NAFNet [48] as the backbone of this module.

At the core of NAFNet are the *NAFBlocks*, as shown in the left bottom Fig. 1, which offer a simplified yet effective alternative to traditional convolutional residual blocks. Each NAFBlock eliminates standard nonlinear activation and normalization layers, replacing them with a lightweight element-wise operation known as *SimpleGate*. For an input feature map $\mathbf{F} \in \mathbb{R}^{H \times W \times 2C}$, SimpleGate operates by first splitting $\mathbf{F}$ along the channel dimension into two halves, $\mathbf{F}_1$ and $\mathbf{F}_2$, each of size $H \times W \times C$. The output is then computed as

$$\text{SimpleGate}\,(\mathbf{F}) \;=\; \mathbf{F}_1 \odot \mathbf{F}_2 \tag{11}$$

where $\odot$ denotes an elementwise multiplication. This operation introduces nonlinearity without the computational overhead of conventional activation functions.

Each NAFBlock combines convolutional layers for feature extraction, the SimpleGate mechanism for nonlinearity, and channel-wise modulation via learned scaling and shifting parameters γ and β

$$\mathbf{F}_{\text{out}} \;=\; \gamma \;\odot\; \text{SimpleGate}\,(\mathbf{F}_{\text{in}}) + \beta \;+\; \mathbf{F}_{\text{in}}. \tag{12}$$

The vectors $\gamma, \beta \in \mathbb{R}^C$ act as feature-wise scale and shift factors. A small multilayer perceptron (MLP) maps the diffusion timestep t to (γ, β), allowing each NAFBlock to dynamically adapt to the current noise level, thereby improving the accuracy of score-function estimation. In addition, both vectors are initialized to zero, which causes the residual path to remain inactive at the beginning of training. This design choice helps stabilize the optimization process, particularly when training very deep NAFNet stacks.

The implementation does not apply clipping to the vectors γ or β. Since they are initialized at zero and regulated by LayerNorm, these parameters typically converge to values within the range of $[-1.5, 1.5]$ for most channels. Occasionally, they may reach up to ± 3, which offers sufficient expressiveness while avoiding the risk of activation explosion.

To enhance temporal conditioning in the diffusion process, the aforementioned MLP produces γ and β from sinusoidal

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time embeddings, granting the network explicit awareness of noise magnitude at every denoising step. An additional lightweight channel-attention branch inside each NAFBlock further enriches global context capture.

Although spatial self-attention is not explicitly present in all layers, the hierarchical U-Net-style structure of NAFNet—supplemented by skip connections between encoder and decoder—facilitates effective multiscale feature fusion. This architecture secures both local texture recovery and global coherence, making the modified NAFBlocks especially suitable for high-resolution dehazing tasks.

To elaborate further, each NAFBlock implicitly acts as a fusion unit where local and global contexts are brought together. Locally, the convolutional operations and SimpleGate mechanisms extract fine-grained features such as textures and edges. Globally, the time-conditioned modulation vectors (γ, β)—derived from sinusoidal embeddings—adaptively guide the network's behavior based on the denoising step, which encodes the broader haze distribution across the scene. Moreover, the embedded lightweight channel attention captures global dependencies at the feature map level. When stacked across multiple resolution levels and paired with encoder–decoder skip connections, NAFBlocks enable seamless integration of multiscale features. This design ensures that fine local details are preserved while being harmonized with global structural information, effectively addressing the spatial nonuniformity inherent in nonhomogeneous haze.

The complete NAFNet, composed of the above blocks, serves as the noise-prediction network within our latent diffusion framework, estimating the noise term $\hat{\boldsymbol{\epsilon}}_\theta$ introduced during forward diffusion and thereby supporting accurate score-function approximation.

The network is trained using a denoising objective that minimizes the difference between the true noise $\boldsymbol{\epsilon}$ and the predicted noise $\hat{\boldsymbol{\epsilon}}_\theta$

$$\mathcal{L} = \mathbb{E}_{\mathbf{z}_0,\boldsymbol{\epsilon},t}\left[\|\boldsymbol{\epsilon} - \hat{\boldsymbol{\epsilon}}_\theta(\mathbf{z}_t, t)\|_2^2\right] \tag{13}$$

where $\mathbf{z}_t$ is obtained from the forward process as

$$\mathbf{z}_t = \boldsymbol{m}_t + \sqrt{v_t} \cdot \boldsymbol{\epsilon} \tag{14}$$

with $\boldsymbol{m}_t$ and v_t denoting the time-dependent mean and variance as derived in (5) and (6), respectively.

To enhance both the discriminative and reconstructive capabilities of latent representations, we employ a latent-replacing training strategy. During training, we encode both clear (high-quality) images and hazy (degraded) images to obtain their respective latent representations. The key step involves replacing the latent representation of the hazy image with that of the clear image before feeding it into the decoder. This replacement guides the decoder to reconstruct the clear image from the clear latent code, effectively teaching the decoder to generate high-quality outputs. By applying reconstruction losses in both the latent space and the image space, the model is encouraged to maintain accurate and meaningful representations, ensuring that the latent space captures essential features necessary for precise image reconstruction. This strategy enhances the decoder's ability to reconstruct clear images from latent codes, thereby improving the overall performance of the dehazing model.

Skip connections linking the encoder and decoder layers facilitate the transfer of feature maps at corresponding resolutions, helping to preserve multiscale features and spatial information throughout the latent diffusion process. This mechanism is essential for maintaining overall structure and coarse-level textures in the dehazed output.

However, operating in the latent space—while computationally efficient—can lead to minor visual degradation, particularly in high-frequency regions. This includes potential artifacts around object boundaries and loss of fine textures, which are critical for perceptual quality in remote sensing imagery. To mitigate these limitations, the encoder and decoder are jointly optimized, and regularization strategies are applied to stabilize training and retain essential content.

Overall, the diffusion-based dehazing phase focuses on removing global haze and restoring structural integrity efficiently. Yet due to the nature of latent-space processing, this stage alone is insufficient for delivering high perceptual quality. Therefore, we introduce a second GAN-based enhancement phase.

B. Enhancement Phase

While the initial dehazing phase of our Dehazing-DiffGAN efficiently recovers the global structure of hazy images, it may introduce residual distortions and noise due to extended training periods and intricate interactions within the latent space of diffusion models. To overcome these challenges and improve the perceptual quality of the dehazed images, we propose an enhancement phase that refines image details and sharpens edges.

This enhancement phase focuses on transforming the output of the dehazing phase, denoted as the low-quality image I^{LQ}, into a high-quality image I^{HQ} that closely approximates the ground-truth image I^{GT}. Our enhancement model is designed to operate in both the spatial domain and the frequency domain, leveraging the strengths of multiscale representations to enhance fine details while preserving structural integrity.

The generator G, parameterized by θ_G, aims to minimize the discrepancy between I^{HQ} and I^{GT}, formulated as

$$\theta_G^* = \arg\min_{\theta_G} \mathbb{E}_{I^{\mathrm{LQ}},I^{\mathrm{GT}}}\left[\mathcal{L}\left(G\left(I^{\mathrm{LQ}};\theta_G\right), I^{\mathrm{GT}}\right)\right] \tag{15}$$

where $\mathcal{L}$ denotes the overall loss function comprising reconstruction, perceptual, adversarial, and Laplacian losses.

The generator G is structured into two primary branches as follows.

1) *Enhancement Branch:* This branch operates in the RGB domain and is dedicated to structural preservation and overall enhancement of I^{LQ}. It utilizes a network architecture based on multiple RRDBs to refine the image by learning residual mappings that correct artifacts and improve visual fidelity. The RRDBs enable the network to effectively capture complex features and patterns necessary for high-quality image reconstruction.
2) *Laplacian Pyramid Branch:* To improve high-frequency details such as edges and textures, we incorporate a

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Laplacian pyramid approach [49], [50]. The Laplacian pyramid serves as a hierarchical representation that breaks down an image into multiple bandpass filtered components, each capturing details at various spatial frequencies. This breakdown enables the network to concentrate on enhancing high-frequency features across multiple scales, which is crucial for restoring fine details lost during the dehazing process.

The construction and utilization of the Laplacian pyramid in our model involve the following steps.

1) *Gaussian Pyramid Generation:* Given an input image I, we create a Gaussian pyramid $I_0, I_1, \ldots, I_N$ by iteratively applying a Gaussian filter and downsampling

$$I_{n+1} = d\left(I_n\right), \quad n = 0, 1, \ldots, N-1 \tag{16}$$

where $d(\cdot)$ represents the downsampling operation using Gaussian blurring. I_0 refers to the original image, and N indicates the number of levels in the pyramid.

2) *Laplacian Pyramid Formation:* The Laplacian pyramid $L_0, L_1, \ldots, L_N$ is constructed by computing the difference between each level of the Gaussian pyramid and an upsampled version of its subsequent level

$$L_n = I_n - u\left(I_{n+1}\right), \quad n = 0, 1, \ldots, N-1 \tag{17}$$

where $u(\cdot)$ denotes the upsampling operation, typically involving interpolation and smoothing to match the resolution of I_n. The last level L_N is set to I_N due to the absence of a subsequent level.

3) *Multiscale Enhancement:* Each Laplacian level L_n captures high-frequency details at scale n. We process these levels through dedicated subnetworks within the Laplacian pyramid branch to enhance the high-frequency components at each scale

$$\hat{L}_n = G_n\left(L_n\right), \quad n = 0, 1, \ldots, N \tag{18}$$

where G_n represents the subnetwork processing level n. These subnetworks are designed to focus on enhancing the specific frequency components present at each scale.

4) *Reconstruction of Enhanced Image:* The enhanced image I^{HQ} is reconstructed from the enhanced Laplacian levels $\hat{L}_0, \hat{L}_1, \ldots, \hat{L}_N$ using the inverse Laplacian pyramid reconstruction

$$\hat{I}_n = u\left(\hat{I}_{n+1}\right) + \hat{L}_n, \quad n = N-1, N-2, \ldots, 0 \tag{19}$$

with $\hat{I}_N = \hat{L}_N$ as the starting point. The final enhanced image is $I^{\mathrm{HQ}} = \hat{I}_0$.

By integrating the Laplacian pyramid into the enhancement phase, our model effectively captures and enhances high-frequency details across multiple scales, leading to improved edge sharpness and texture fidelity in the final dehazed images. This multiscale approach allows the network to focus on both global structure and local details, resulting in a more comprehensive enhancement.

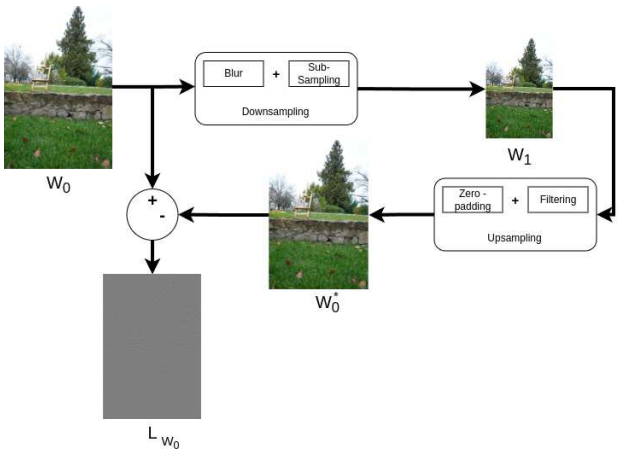


Fig. 2. Illustration of the Laplacian Pyramid decomposition and reconstruction process. The original image W_0 is decomposed into different detail levels by applying downsampling and upsampling operations, leading to the extraction of high-frequency components, represented as L_{W_0}. These components are enhanced and recombined to reconstruct the image with improved clarity.

1) Visualization of the Laplacian Pyramid Process: Fig. 2 illustrates the process of decomposing an image into its high-frequency components. The input image W_0 undergoes a downsampling operation, resulting in a lower resolution version W_1, which captures the coarser details of the scene. To isolate the finer details, the image is upsampled and filtered, resulting in an estimated image W_0^* that is subtracted from W_0 to extract the Laplacian layer L_{W_0}, which emphasizes edges and textures.

This process showcases how the Laplacian decomposition captures essential visual features, making it a powerful tool for tasks like image enhancement and dehazing. By isolating high-frequency details and later enhancing them, we achieve a sharper and more detailed reconstruction of the image.

The combination of decomposition and enhancement, as depicted in Fig. 2, highlights the strengths of multistage processing techniques. It enables the model to focus on fine details that contribute significantly to image clarity, producing results that are both visually enhanced and faithful to the original scene.

2) Loss Functions: To train the enhancement phase effectively, we define a comprehensive loss function that combines multiple objectives as follows.

1) *Reconstruction Loss:* An L_1 loss between the enhanced image I^{HQ} and the ground-truth image I^{GT}

$$\mathcal{L}_{\mathrm{rec}} = |I^{\mathrm{HQ}} - I^{\mathrm{GT}}|_1. \tag{20}$$

2) *Perceptual Loss:* A perceptual loss computed using feature maps from a pretrained network (e.g., VGG) to measure the similarity of high-level representations

$$\mathcal{L}_{\mathrm{per}} = \sum_l \left\|\phi_l\left(I^{\mathrm{HQ}}\right) - \phi_l\left(I^{\mathrm{GT}}\right)\right\|_2^2 \tag{21}$$

where $\phi_l(\cdot)$ denotes the activation of layer l.

3) *Adversarial Loss:* An adversarial loss based on a discriminator D that encourages the generator to produce

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images indistinguishable from real images. Formally, it is defined as

$$\mathcal{L}_{\text{adv}} = -\mathbb{E}_{I^{\text{HQ}}}\left[\log D\left(I^{\text{HQ}}\right)\right] \tag{22}$$

where $\mathbb{E}$ denotes the expectation operator, i.e., the average value over a batch of generated images I^{HQ}. The negative sign "-" is included because, in the original GAN formulation, the discriminator aims to maximize the log-likelihood of correctly classifying real versus generated images.

4) *Laplacian Loss:* A loss that specifically measures the difference between the Laplacian pyramids of I^{HQ} and I^{GT} to ensure high-frequency details are accurately reconstructed

$$\mathcal{L}_{\text{lap}} = \sum_{n=0}^{N-1} \left\| \hat{L}_n - L_n^{\text{GT}} \right\|_1 \tag{23}$$

where L_n^{GT} denotes the nth level of the Laplacian pyramid of the ground-truth image, $\hat{L}_n$ is the corresponding level of the predicted image, and N is the total number of pyramid levels used. The total loss function is then defined as

$$\mathcal{L} = \lambda_{\text{rec}}\mathcal{L}_{\text{rec}} + \lambda_{\text{per}}\mathcal{L}_{\text{per}} + \lambda_{\text{adv}}\mathcal{L}_{\text{adv}} + \lambda_{\text{lap}}\mathcal{L}_{\text{lap}} \tag{24}$$

where λ_{rec}, λ_{per}, λ_{adv}, and λ_{lap} are weighting factors balancing the contributions of each loss component. To ensure effective joint training of the enhancement phase, we adopt a balanced loss configuration aligned with established practices in prior works such as [15] and [51]. Specifically, we set the reconstruction loss coefficient to $\lambda_{\text{rec}} = 0.01$, the perceptual (VGG) loss to $\lambda_{\text{per}} = 1.0$, and the adversarial loss to $\lambda_{\text{adv}} = 0.005$. For the Laplacian-consistency loss, we perform a targeted grid search over $\lambda_{\text{lap}} \in \{0.01, 0.02, 0.05\}$ using the SateHaze1k validation set. The average PSNR values obtained for these weights are 25.23, 23.61, and 24.68 dB, respectively. Similar rankings were observed for SSIM and LPIPS metrics. Based on these results, we select $\lambda_{\text{lap}} = 0.01$ as it provides the best tradeoff between reconstruction accuracy and perceptual quality, and we use this setting in all subsequent experiments.

By incorporating the enhancement phase into the Dehazing-DiffGAN framework, we effectively address the residual artifacts from the initial dehazing phase. The use of the Laplacian pyramid allows for meticulous enhancement of details across multiple scales, resulting in superior edge sharpness and texture reproduction. This comprehensive approach leads to high-quality dehazed images suitable for real-world applications.

Beyond the tailored loss formulation, *Dehazing-DiffGAN* employs a **maximum-ensemble strategy** (see Fig. 3) to refine model weights. Every ten epochs, we measure performance with the LPIPS metric—widely acknowledged for capturing human-perceived image quality—and keep only those checkpoints that outperform the current best. To ensure transparency, LPIPS values at each ten-epoch interval are explicitly tracked

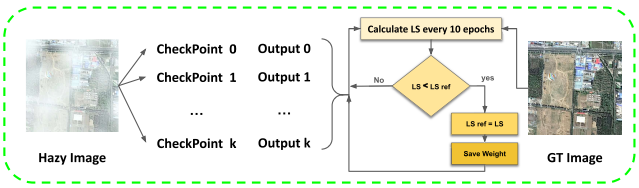


Fig. 3. Maximum checkpoint ensemble method.

TABLE II

PERFORMANCE COMPARISON OF DEHAZING MODELS IN THE NTIRE 2024 COMPETITION. OUR MODEL ACHIEVED THE BEST LPIPS SCORE, WITH THE BEST RESULTS HIGHLIGHTED IN BOLD

Model	PSNR↑	SSIM↑	LPIPS↓	Params.(M)
DH-AISP	21.902	0.714	0.401	1418
Team Woof	22.59	0.726	0.380	521
USTC-Dehazers	**22.943**	**0.729**	0.352	150.16
Dehazing R	22.84	0.725	0.346	256
TTWT	21.932	0.714	0.334	-
ITB Dehaze	22.323	0.714	0.333	110
RepD	21.782	0.706	0.322	41.32
BU-Dehaze	21.676	0.709	0.326	13.4
Dehazing-DiffGAN (our)	20.538	0.632	**0.267**	161

TABLE III

COMPARISON OF THE PROPOSED DEHAZING-DIFFGAN WITH STATE-OF-THE-ART METHODS INCLUDING REFUSSION, SDM, SGLC, DWT-FFC-GAN, AND 4K-DEHAZER. THE BEST RESULTS FOR EACH METRIC ARE MARKED IN BOLD

Method	Year	Performance Metrics		
		PSNR↑	SSIM↑	LPIPS↓
4K-Dehazer [7]	2021	12.727696	0.629890	0.501968
SDM [35]	2022	20.851901	0.646988	0.338828
Refussion [36]	2023	23.401497	0.741354	0.319131
SGLC [28]	2023	**23.432630**	**0.748979**	0.353761
DWT-FFC-GAN [25]	2023	21.749818	0.744119	0.306642
Dehazing-DiffGAN (our)	-	23.069909	0.733270	**0.294703**

to observe whether performance improves or plateaus throughout training. Prioritizing perceptual fidelity over raw pixel agreement markedly boosts restoration quality.

Concretely, let $\mathbf{I}_{\text{hazy}}$ be the input. During training, n checkpoints yield outputs $\{\mathbf{O}_0, \mathbf{O}_1, \ldots, \mathbf{O}_n\}$, where $\mathbf{O}_i$ corresponds to the weights stored at the ith checkpoint. The **maximum-ensemble rule** forms the final prediction $\mathbf{I}_{\text{clean}}$ by selecting, for each pixel, the checkpoint output with the highest perceptual similarity

$$\mathbf{I}_{\text{clean}} = \arg\max_{i\in\{0,\ldots,n\}} S\left(\mathbf{I}_{\text{hazy}}, \mathbf{O}_i\right) \tag{25}$$

where $S(\cdot,\cdot)$ is a similarity measure (LPIPS in practice).

Because the LPIPS test is run every ten epochs, only the superior checkpoints $\{\mathbf{W}_0, \mathbf{W}_1, \ldots, \mathbf{W}_n\}$ are retained, and their outputs are fused

$$\mathbf{I}_{\text{clean}} = \text{Ensemble}\left(\mathbf{O}_0, \mathbf{O}_1, \ldots, \mathbf{O}_n\right) \tag{26}$$

with `Ensemble` applying the rule in (25).

This disciplined checkpointing-ensemble scheme cuts down on manual validation while delivering a more robust and higher quality dehazing result, as it harnesses the strongest weights found at different stages of training under varied haze conditions.

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TABLE IV

PERFORMANCE COMPARISON OF THE PROPOSED DEHAZING-DIFFGAN WITH STATE-OF-THE-ART DEHAZING MODELS UNDER VARYING FOG CONDITIONS (THIN, MODERATE, AND THICK FOG) USING PSNR, SSIM, AND LPIPS METRICS. THIS TABLE HIGHLIGHTS MODELS EVALUATED ON LPIPS IN ADDITION TO PSNR AND SSIM. THE BEST RESULTS FOR EACH CONDITION ARE SHOWN IN BOLD

Model	Year	Thin Fog			Moderate Fog			Thick Fog			Average		
		PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
DCP [5]	2010	15.99	0.835	0.10	14.75	0.821	0.13	10.99	0.617	0.27	13.91	0.758	0.17
AOD-Net [52]	2017	18.47	0.863	0.08	17.63	0.855	0.12	15.75	0.742	0.23	17.28	0.820	0.14
FCTF-Net [53]	2020	19.71	0.875	0.10	23.10	0.926	0.06	18.56	0.800	0.20	20.46	0.867	0.12
PFF-Net [54]	2019	16.01	0.821	0.17	18.59	0.688	0.49	16.06	0.575	0.61	16.89	0.695	0.42
GridDehaze-Net [55]	2019	19.36	0.857	0.09	21.91	0.905	0.07	17.83	0.773	0.20	19.70	0.845	0.12
FFA-Net [37]	2020	24.26	0.910	0.06	25.39	0.930	0.08	21.83	0.836	0.16	23.83	0.892	0.10
SCA-Net [38]	2023	19.70	0.882	0.06	24.75	0.934	0.05	18.40	0.812	0.13	20.95	0.876	0.08
AIDTransformer [56]	2023	24.82	0.904	0.048	27.20	0.918	0.039	22.21	0.831	0.107	24.74	0.884	0.065
EDED-Net [39]	2024	24.81	**0.924**	0.05	25.65	**0.939**	0.05	22.46	0.857	0.13	24.31	**0.907**	0.08
PGSformer [57]	2024	25.53	0.9176	0.0660	26.62	0.9328	0.0665	23.59	**0.8628**	0.1378	25.25	0.9044	0.0901
Dehazing-DiffGAN (our)	-	**27.084**	0.8863	**0.0413**	**30.09**	0.9247	**0.0315**	**24.45**	0.8125	**0.0689**	**27.21**	0.8745	**0.0472**

TABLE V

PERFORMANCE COMPARISON OF THE PROPOSED DEHAZING-DIFFGAN WITH STATE-OF-THE-ART DEHAZING MODELS UNDER VARYING FOG CONDITIONS (THIN, MODERATE, AND THICK FOG) USING PSNR AND SSIM METRICS ONLY. ADDITIONAL PARAMETERS ARE INCLUDED FOR COMPLETENESS. THE BEST RESULTS FOR EACH CONDITION ARE SHOWN IN BOLD

Model	Year	Parameters (M)	Thin Fog		Moderate Fog		Thick Fog		Average	
			PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑
cGAN [40]	2020	-	24.164	0.906	25.311	0.9264	25.073	0.864	24.849	0.8988
UMWTransformer [58]	2022	4.5	24.29	0.919	26.65	0.946	20.07	0.825	23.67	0.8967
FocalNet [59]	2023	**3.74**	24.16	0.916	25.99	0.947	21.69	0.847	23.947	0.9033
PSMB-Net [60]	2023	12.45	26.75	0.928	27.48	0.946	25.15	0.889	26.46	0.921
ARDD-Net [61]	2023	-	26.84	0.9257	26.47	0.9321	26.83	0.9316	26.71	0.930
ConvIR-S [42]	2024	14.83	25.11	**0.978**	26.79	**0.978**	22.65	**0.950**	24.85	**0.9687**
ADND-Net [45]	2024	160.14	26.91	0.9274	26.67	0.9358	**26.94**	0.9358	26.84	0.933
PhDnet [43]	2024	10.03	23.13	0.8969	26.47	0.9421	19.251	0.8069	22.95	0.882
HSMD-Net [44]	2024	9.91	26.91	0.928	28.07	0.957	24.81	0.88	26.597	0.919
SCSNet [62]	2024	-	26.146	0.9415	28.350	0.9566	24.654	0.9015	26.383	0.936
RSDehamba [63]	2024	1.80	26.750	0.9306	27.45	0.9468	23.53	0.8698	25.91	0.9157
Dehazing-DiffGAN (ours)	-	161	**27.08**	0.8863	**30.09**	0.9247	24.45	0.8125	**27.21**	0.8745

IV. RESULTS AND DISCUSSION

A. Experimental Settings

In this study, we thoroughly analyzed the performance of our proposed *Dehazing-DiffGAN* against over 15 advanced dehazing methods, including prominent models, such as DCP [5], AOD-Net [52], FCTF-Net [53], PFF-Net [54], GridDehaze-Net [55], FFA-Net [37], SCA-Net [38], AID-Transformer [56], EDED-Net [39], ConvIR-S [42], ADND-Net [45], PhDnet [43], HSMD-Net [44], among others. Tables II and III present comparative evaluations on generic haze datasets, highlighting both perceptual and quantitative results across multiple metrics. To further assess the robustness of our approach under varying real-world conditions, Tables IV and V report detailed results on remote sensing datasets with different levels of haze intensity, demonstrating the effectiveness of Dehazing-DiffGAN in challenging atmospheric environments.

Our comparative analysis involves evaluating a series of dehazing models on two specialized remote sensing datasets. To gauge the efficacy of these models, we employed three primary metrics: PSNR, SSIM, and LPIPS.

PSNR evaluates the ratio between the peak power of a signal and the corrupting noise, offering a measure of the reconstructed image's fidelity. Higher PSNR values indicate improved reconstruction quality with reduced noise. SSIM assesses image quality by measuring the structural similarities between the original and dehazed images, focusing on parameters such as luminance, contrast, and structure.

LPIPS, on the other hand, utilizes deep learning to approximate human visual perception by comparing feature representations of images from a neural network. Lower LPIPS values indicate a higher perceptual resemblance to the original images, making this metric particularly important for assessing fine textural fidelity and overall visual quality.

These metrics were calculated by averaging the performance across all image channels, ensuring a comprehensive assessment of each model's ability to restore clarity and detail in dehazed images.

The mathematical formulations for these metrics are as follows:

$$\text{PSNR} = 10 \cdot \log_{10}\left(\frac{\text{MAX}^2}{\text{mse}}\right) \tag{27}$$

where MAX represents the maximum possible pixel value, and mse is the mean squared error between the original and reconstructed images.

The SSIM index is determined by

$$\text{SSIM}(x, y) = \frac{\left(2\mu_x\mu_y + c_1\right)\left(2\sigma_{xy} + c_2\right)}{\left(\mu_x^2 + \mu_y^2 + c_1\right)\left(\sigma_x^2 + \sigma_y^2 + c_2\right)} \tag{28}$$

where μ_x and μ_y are the means, σ_x^2 and σ_y^2 are the variances, and σ_{xy} is the covariance of the windows x and y. The con-

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stants c_1 and c_2 are used to stabilize the division, particularly when the denominator is close to zero.

LPIPS is calculated using

$$\text{LPIPS}(x, y) = \sum_l w_l \cdot \frac{1}{H_l W_l} \sum_{h,w} \left\| \phi_l(x)_{h,w} - \phi_l(y)_{h,w} \right\|_2^2 \quad (29)$$

where $\phi_l(x)$ and $\phi_l(y)$ represent the feature maps at layer l, with H_l and W_l denoting the height and width of these feature maps. The weights w_l are layer-specific and emphasize the contribution of each layer to the overall perceptual dissimilarity.

Dehazing-DiffGAN was implemented using PyTorch 2.3.1 and CUDA 11.8, and it was optimized to run on an NVIDIA QUADRO 8000 GPU. The network architecture consists of two primary stages. In the dehazing phase, the model operates on full-resolution images, applying latent space transformations to remove haze. The enhancement phase refines the dehazed images using a GAN-based architecture. The model was evaluated on two datasets: DNH-HAZE, which consists of generic images with a resolution of 6000×4000, and SateHaze1k, containing remote sensing images at a resolution of 512×512.

To account for variability in training due to different dataset sizes, one-third of the dataset is randomly sampled in each epoch, resulting in a total of 1000 epochs. The first 50 epochs serve as a warm-up phase, while the final 200 epochs implement the frozen batch normalization (FrozenBN) technique to enhance training stability. The model is trained using a batch size of 8, beginning with an initial learning rate of 1×10^{-4}, and optimized with the Adam optimizer ($\beta_1 = 0.9, \beta_2 = 0.999$). A cosine annealing schedule is applied, progressively lowering the learning rate to 1×10^{-6} by the conclusion of the training process.

B. Datasets

To assess the effectiveness of our dehazing methods, we employed two different datasets: the DNH-HAZE dataset provided by the NTIRE 2024 Image Dehazing Challenge and the SateHaze1k dataset, which is specifically tailored for remote sensing.

1) DNH-HAZE Dataset: The DNH-HAZE dataset was introduced for the NTIRE 2024 Image Dehazing Challenge and builds on prior datasets such as NH-Haze and HD-NH-HAZE. It includes 50 image pairs, each featuring a hazy image along with its corresponding clear version of the same outdoor scene. The haze conditions are dense and nonhomogeneous, created with professional haze machines that generate vapor particles between 1 and 10 μm in diameter, closely mimicking natural atmospheric haze particles. High-resolution images were captured using remotely controlled Sony A7 III cameras. Consistent camera parameters, such as shutter speed, aperture, ISO, and white balance, were maintained during the recording sessions to ensure precise pixel-level alignment between image pairs. The DNH-HAZE dataset provides a realistic and challenging benchmark for assessing dehazing algorithms under complex haze conditions.

2) SateHaze1k Dataset: Introduced by Huang et al. [40], the SateHaze1k dataset is tailored for remote sensing applications and divided into three subsets based on haze intensity: SateHaze1k-thick, SateHaze1k-moderate, and SateHaze1k-thin. Each subset comprises 320 RGB images designated for training and 45 RGB images for testing. The "thin" haze subset simulates light fog conditions using haze masks derived from actual cloud images. The "moderate" subset combines attributes of mist and moderate fog, presenting a medium level of haze complexity. The "thick" haze subset depicts the most challenging scenarios by applying transmittance maps from highly hazed environments to replicate dense fog conditions. Collectively, these subsets enable a thorough evaluation of dehazing algorithms across various levels of difficulty.

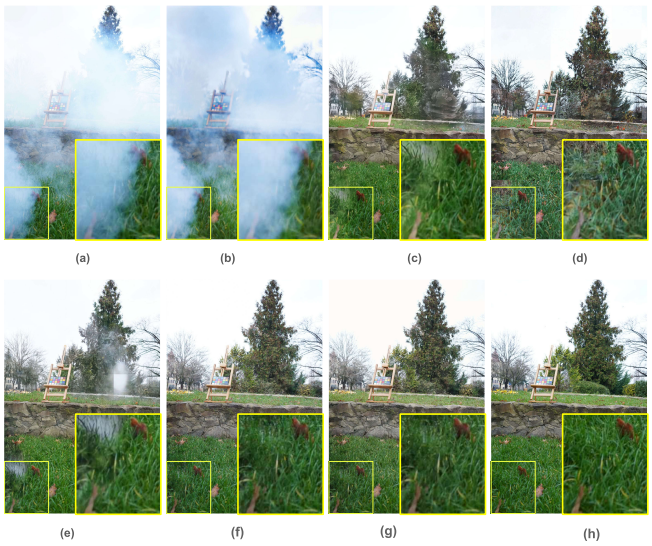


Fig. 4. Visual comparison of dehazing results from the proposed Dehazing-DiffGAN and several state-of-the-art methods (4K-Dehazer, SGLC, SDM, and DWT-FFC-GAN) on the DNH-HAZE dataset. The figure includes: (a) input hazy image, (b) 4K-Dehazer, (c) SLGE, (d) SDM, (e) DWT-FFC-GAN, (f) output from the dehazing phase of Dehazing-DiffGAN, (g) final output of Dehazing-DiffGAN, and (h) GT.

These datasets are crucial for our experimental validation, offering diverse and challenging conditions that closely mirror real-world generic and remote sensing scenarios. Their inclusion ensures the robustness and applicability of our dehazing solutions across a wide spectrum of practical environments.

C. Performance Evaluation Across Diverse Haze Conditions

To rigorously evaluate the performance of *Dehazing-DiffGAN*, we methodically divided our analysis into two distinct domains, each designed to demonstrate the model's adaptability and precision under varied operational scenarios. The first domain leverages the DNH-HAZE dataset to assess the model's efficiency in handling generic outdoor scenes characterized by dense and nonhomogeneous haze distributions. In contrast, the second domain examines the model's capability to effectively mitigate atmospheric haze across a broad spectrum of densities—ranging from thin to thick—using the SateHaze1k dataset.

1) Haze Removal Performance for Dense and Nonhomogeneous Haze: In this section, we evaluate the performance of our proposed *Dehazing-DiffGAN* on the NTIRE 2024 Dense and Nonhomogeneous Haze Removal Challenge dataset, as

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shown in Fig. 4. We compare the results of our method with several state-of-the-art dehazing models using key metrics such as PSNR, SSIM, and LPIPS.

As shown in Table II, our proposed Dehazing-DiffGAN model achieved the best performance in terms of perceptual quality, as evidenced by the LPIPS score. Specifically, the Dehazing-DiffGAN model attained an LPIPS score of **0.267**, outperforming all other competing models. In comparison, the second-best LPIPS score was achieved by the RepD model, with a value of 0.322. The reduction in LPIPS indicates that our model is particularly effective at restoring high-quality perceptual features, leading to visually appealing dehazed images.

Although the Dehazing-DiffGAN model did not achieve the highest PSNR or SSIM scores, it is important to note that the primary goal in dehazing dense and nonhomogeneous scenes is often to enhance perceptual quality rather than focusing solely on pixel-level fidelity. For instance, the USTC-Dehazers model achieved the highest PSNR and SSIM scores at **22.943** and **0.729**, respectively, yet its LPIPS score of 0.352 reflects a perceptual quality lower than that of Dehazing-DiffGAN.

Table III provides further evidence of the efficacy of our approach, showing a comparison of our results with those of top-performing models from previous NTIRE competitions on the same dataset. Our model outperforms recent state-of-the-art methods such as Refussion [36], SGLC [28], and DWT-FFC-GAN [25] in terms of LPIPS, achieving the best score of **0.294**, while maintaining competitive PSNR and SSIM values.

The superiority of Dehazing-DiffGAN in perceptual quality can also be observed in Fig. 4, where visual comparisons clearly demonstrate the enhanced clarity and reduced haze artifacts in images processed by our model. This improvement can be attributed to the dual-phase architecture of our approach, where the initial dehazing is performed by a diffusion-based model, followed by a GAN-based enhancement that refines high-frequency details using a Laplacian Pyramid approach. This dual-phase process ensures both global haze removal and local detail preservation, making our model highly effective in challenging dehazing scenarios.

2) Performance Evaluation of Haze Removal Across Varying Haze Intensities in Remote Sensing: We evaluate the performance of the proposed *Dehazing-DiffGAN* model for haze removal in remote sensing images across different levels of haze intensity using the *SateHaze1k* dataset. The dataset includes three distinct haze conditions: thin, moderate, and thick. We compare our method against several state-of-the-art dehazing models using PSNR, SSIM, and LPIPS metrics.

Table IV presents a detailed quantitative comparison of the performance of different models under thin, moderate, and thick haze conditions. Our proposed *Dehazing-DiffGAN* model consistently outperforms competing methods across all haze intensities. For thin haze conditions, *Dehazing-DiffGAN* achieves a PSNR of **27.084**, surpassing other leading models such as *EDED-Net* (PSNR 24.81) and *FFA-Net* (PSNR 24.26). In the case of moderate haze, *Dehazing-DiffGAN* registers the highest PSNR of **30.09**, outperforming *EDED-Net* (PSNR 25.65) and *SCA-Net* (PSNR 24.75). Even under thick haze, which poses significant challenges, *Dehazing-DiffGAN* maintains the best PSNR of **24.45**, illustrating its robustness in handling severe haze.

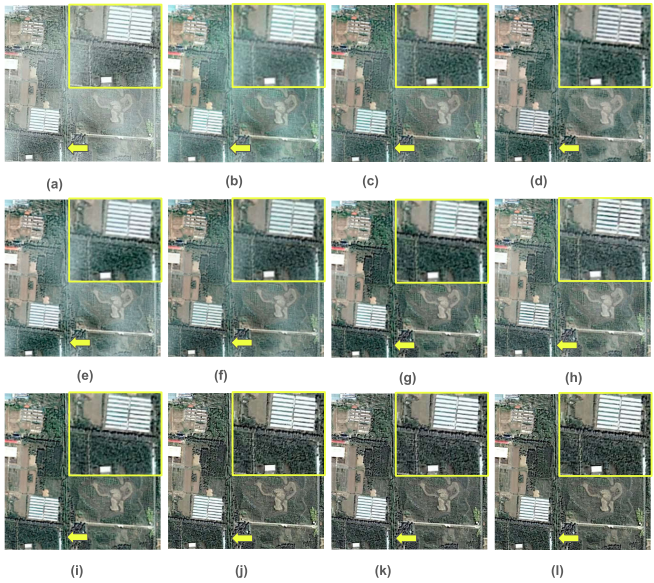


Fig. 5. Comparative visualizations based on the SateHaze1k-thin dataset: (a) original hazy image; (b) DCP [5]; (c) AOD-Net [52]; (d) FCTF-Net [53]; (e) PFF-Net [54]; (f) GridDehaze-Net [55]; (g) FFA-Net [37]; (h) SCA-Net [38]; (i) EDED-Net [39]; (j) our *dehazing phase*; (k) our proposed *Dehazing-DiffGAN*; and (l) ground truth image. All images from (b) to (k) have been processed using their respective dehazing methods.

Notably, *Dehazing-DiffGAN* achieves the highest LPIPS scores across all fog intensities, demonstrating superior perceptual quality in the restored images when compared to other advanced models. The LPIPS metric, which evaluates perceptual similarity using deep network features, further underscores the effectiveness of *Dehazing-DiffGAN* in producing dehazed images that closely resemble the ground truth.

As indicated by the average results across all haze conditions, shown in Table IV, the *Dehazing-DiffGAN* model achieves an average PSNR of **27.21** and LPIPS of **0.0472**, highlighting its effectiveness in dehazing a diverse range of remote sensing images. The exceptional performance of *Dehazing-DiffGAN* across these quantitative metrics positions it as a leading model for remote sensing image dehazing.

Table V further compares our method against other state-of-the-art models using only PSNR and SSIM metrics. Once again, *Dehazing-DiffGAN* demonstrates its robustness, achieving the best average PSNR of **27.21** across thin, moderate, and thick haze conditions. These results confirm the capability of *Dehazing-DiffGAN* to handle varying haze intensities effectively, providing consistent and high-quality dehazing outcomes.

In addition to the quantitative assessment, we provide qualitative visual comparisons for different fog intensities using the *SateHaze1k* dataset. Fig. 5 illustrates the results for thin haze conditions, where the proposed *Dehazing-DiffGAN* model delivers clearer, more visually appealing images with sharper details compared to competing methods. Similarly, Fig. 6 shows the visual performance under moderate haze conditions, where *Dehazing-DiffGAN* again outperforms other models in terms of haze removal and detail restoration. For

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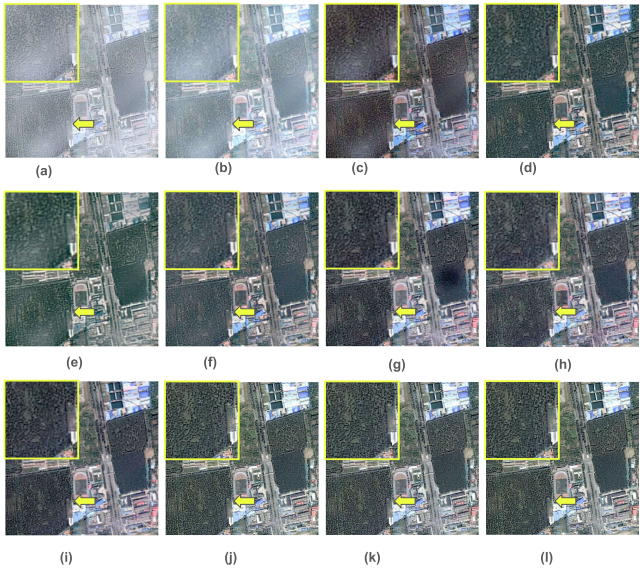


Fig. 6. Comparative visualizations based on the SateHaze1k-moderate dataset: (a) original hazy image; (b) DCP [5]; (c) AOD-Net [52]; (d) FCTF-Net [53]; (e) PFF-Net [54]; (f) GridDehaze-Net [55]; (g) FFA-Net [37]; (h) SCA-Net [38]; (i) EDED-Net [39]; (j) our *dehazing phase*; (k) our proposed *Dehazing-DiffGAN*; and (l) ground truth image. All images from (b) to (k) have been processed using their respective dehazing methods.

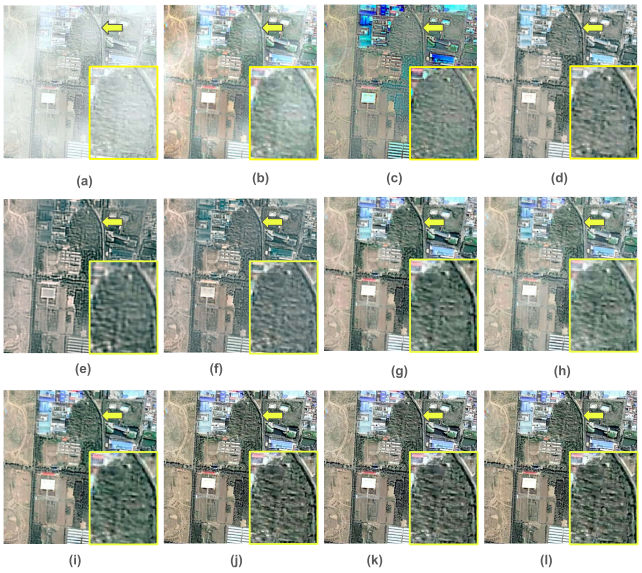


Fig. 7. Comparative visualizations based on the SateHaze1k-thick dataset: (a) original hazy image; (b) DCP [5]; (c) AOD-Net [52]; (d) FCTF-Net [53]; (e) PFF-Net [54]; (f) GridDehaze-Net [55]; (g) FFA-Net [37]; (h) SCA-Net [38]; (i) EDED-Net [39]; (j) our *dehazing phase*; (k) our proposed *Dehazing-DiffGAN*; and (l) ground truth image. All images from (b) to (k) have been processed using their respective dehazing methods.

thick haze, Fig. 7 shows that *Dehazing-DiffGAN* excels in handling even the most challenging haze scenarios, producing dehazed images that are perceptually superior in terms of texture and color fidelity.

In real-world remote sensing applications, perceptual quality is critical, particularly when the images are used for downstream tasks such as land-cover classification, object detection, or urban monitoring. A lower LPIPS score reflects perceptual closeness to ground truth, indicating better preservation of textures and visual realism. As observed in Figs. 4–7, the improvements in LPIPS translate to clearer edges, more natural textures (e.g., vegetation and rooftops), and fewer artifacts—especially in high-frequency regions. These enhancements make the output more suitable for both human interpretation and machine-driven analysis, highlighting the practical relevance of perceptual gains.

D. Ablation Study

To further analyze the effectiveness of the proposed *Dehazing-DiffGAN* model, we conducted an ablation study, where we compared the full model with an intermediate version, referred to as *dehazing phase*, and the hazy input image under varying fog intensities, as shown in Table VI. The comparison was performed across three different fog conditions: thin, moderate, and thick haze, using three metrics—PSNR, SSIM, and LPIPS.

From the results in Table VI, it is clear that both *dehazing phase* and the full *Dehazing-DiffGAN* model outperform the hazy input across all evaluated metrics and haze conditions. For thin fog, the *Dehazing-DiffGAN* model achieves a PSNR of **27.084** and an SSIM of **0.8863**, reflecting slight enhancements over *dehazing phase* (PSNR of 26.751 and SSIM of 0.8793). The LPIPS score of **0.0413** for the full model is highly competitive relative to *dehazing phase*'s LPIPS of 0.0400, demonstrating superior perceptual quality in dehazed results.

Under moderate fog conditions, the full model's PSNR of **30.094** demonstrates superior performance compared to *dehazing phase* (PSNR of 30.346), although the SSIM of **0.9247** is marginally lower. The LPIPS of **0.0315** remains competitive, confirming the perceptual quality of the generated images.

For thick fog, the *Dehazing-DiffGAN* model maintains a PSNR of **24.451** and an SSIM of **0.8125**, outperforming *dehazing phase* (PSNR of 24.128 and SSIM of 0.8024) and showing robustness in handling severe fog. The LPIPS of **0.0689** remains close to that of *dehazing phase*, further indicating the strength of the proposed model.

On average, across all fog conditions, the *Dehazing-DiffGAN* model achieves the highest overall PSNR of **27.209** and an SSIM of **0.8745**, with a competitive LPIPS score of **0.0472**. These results indicate that the dual-phase architecture of the *Dehazing-DiffGAN* model is crucial for effectively removing haze and enhancing image quality in remote sensing applications, particularly in challenging environments with dense haze.

Despite the strong performance of the *Dehazing-DiffGAN* model, several limitations must be acknowledged. In particular, under extreme haze conditions such as thick fog, the model shows slight degradation in perceptual quality, as indicated by a relatively higher LPIPS score in Table VI. This suggests difficulty in accurately recovering fine textures when structural cues are heavily obscured. Visual results in Fig. 7 further illustrate minor residual artifacts in complex regions.

In addition, the computational complexity of the model—comprising high parameters and a dual-phase pipeline—limits its suitability for real-time applications. This constraint is especially relevant in scenarios requiring

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TABLE VI

ABLATION STUDY RESULTS OF OUR *Dehazing – DiffGAN* MODEL COMPARED TO DEHAZING PHASE AND THE HAZY INPUT UNDER DIFFERENT FOG CONDITIONS

Model	Thin Fog			Moderate Fog			Thick Fog			Average		
	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
Hazy	14.200	0.7486	0.1672	14.298	0.7683	0.1741	10.009	0.4519	0.4259	12.836	0.6563	0.2557
Dehazing Phase	26.751	0.8793	0.0400	30.346	0.9281	0.0275	24.128	0.8024	0.0683	27.075	0.8699	0.0453
Dehazing-DiffGAN (our)	27.084	0.8863	0.0413	30.094	0.9247	0.0315	24.451	0.8125	0.0689	27.209	0.8745	0.0472

on-device or low-latency processing. Future work will explore model compression, architectural simplification, and the development of lightweight variants to enhance deployment efficiency without compromising visual quality. Further improvements may also include the incorporation of depth priors, uncertainty modeling, or multimodal auxiliary data (e.g., depth estimation and haze transmission maps) to better guide reconstruction in severely degraded scenes. Another potential direction is to incorporate uncertainty-aware or attention-driven mechanisms that explicitly focus on ambiguous or heavily degraded regions.

From a practical standpoint, Dehazing-DiffGAN is well-suited for high-fidelity remote sensing applications where restoration quality is prioritized over inference speed—such as land-use classification, postdisaster analysis, and environmental monitoring. However, for real-time or edge deployment scenarios (e.g., UAVs or onboard satellite systems), further optimization is required. Future work will aim to bridge this gap by developing compressed or lightweight variants that preserve perceptual performance while significantly reducing computational overhead.

V. CONCLUSION

In this study, we introduced *Dehazing-DiffGAN*, a novel dual-phase deep learning framework that combines diffusion models with GANs for effective image dehazing. The dehazing phase uses an encoder–decoder structure with NAFBlocks to capture both local and global context efficiently. The enhancement phase applies a dual-domain enhancement technique that refines the dehazed images in both RGB and Laplacian domains, allowing better preservation of low-frequency content and fine edge details. Extensive experiments on challenging datasets, including the NTIRE nonhomogeneous dehazing dataset and the remote sensing-focused SateHaze1k dataset, demonstrate that *Dehazing-DiffGAN* consistently outperforms existing methods in terms of PSNR, SSIM, and LPIPS. These results highlight the robustness and versatility of our approach, making it well-suited for a wide range of image dehazing tasks, including remote sensing applications.

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REFERENCES

[1] J. Yang, Z. Wang, D. Chen, S. Xiao, and A. Taher Azar, "A joint network of super-resolution and active learning for agricultural land classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 13669–13677, 2024.

[2] A. Jabbar et al., "Deep transfer learning-based automated diabetic retinopathy detection using retinal fundus images in remote areas," *Int. J. Comput. Intell. Syst.*, vol. 17, no. 1, p. 135, May 2024.

[3] M. Zakzouk, I. A. El-Magd, E. M. Ali, A. M. Abdulaziz, A. Rehman, and T. Saba, "Novel oil spill indices for sentinel-2 imagery: A case study of natural seepage in Qaruh Island, Kuwait," *MethodsX*, vol. 12, Jun. 2024, Art. no. 102520.

[4] Q. Zhu, J. Mai, and L. Shao, "A fast single image haze removal algorithm using color attenuation prior," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 3522–3533, Nov. 2015.

[5] K. He, J. Sun, and X. Tang, "Single image haze removal using dark channel prior," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 12, pp. 2341–2353, Dec. 2011.

[6] E. Akleman, "Deep learning," *Computer*, vol. 53, no. 9, pp. 17–17, Sep. 2020.

[7] Z. Zheng et al., "Ultra-high-definition image dehazing via multi-guided bilateral learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 16180–16189.

[8] C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 2, pp. 295–307, Feb. 2016.

[9] K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang, "Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising," *IEEE Trans. Image Process.*, vol. 26, no. 7, pp. 3142–3155, Jul. 2017.

[10] X. Tao, H. Gao, X. Shen, J. Wang, and J. Jia, "Scale-recurrent network for deep image deblurring," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8174–8182.

[11] W. Ren, S. Liu, H. Zhang, J. Pan, X. Cao, and M. Yang, "Single image dehazing via multi-scale convolutional neural networks," in *Proc. ECCV*, Oct. 2016, pp. 154–169.

[12] D. Ren, W. Zuo, Q. Hu, P. Zhu, and D. Meng, "Progressive image deraining networks: A better and simpler baseline," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3937–3946.

[13] C. Ledig et al., "Photo-realistic single image super-resolution using a generative adversarial network," in *Proc. CVPR*, Jun. 2017, pp. 105–114.

[14] M. S. M. Sajjadi, B. Scholkopf, and M. Hirsch, "EnhanceNet: Single image super-resolution through automated texture synthesis," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4491–4500.

[15] X. Wang et al., "ESRGAN: Enhanced super-resolution generative adversarial networks," in *Proc. ECCVW*, 2019, pp. 63–79.

[16] J. W. Soh, G. Y. Park, J. Jo, and N. I. Cho, "Natural and realistic single image super-resolution with explicit natural manifold discrimination," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 8114–8123.

[17] W. Wang, R. Guo, Y. Tian, and W. Yang, "CFSNet: Toward a controllable feature space for image restoration," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4140–4149.

[18] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 5998–6008.

[19] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[20] X. Chen, X. Wang, J. Zhou, Y. Qiao, and C. Dong, "Activating more pixels in image super-resolution transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 22367–22377.

[21] Z. Tu et al., "MaxViT: Multi-axis vision transformer," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 459–479.

[22] C. Dong, C. C. Loy, and X. Tang, "Accelerating the super-resolution convolutional neural network," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 391–407.

[23] Y. Zhang, Y. Tian, Y. Kong, B. Zhong, and Y. Fu, "Residual dense network for image super-resolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2472–2481.

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[24] J. Kim, J. K. Lee, and K. M. Lee, "Accurate image super-resolution using very deep convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1646–1654.

[25] H. Zhou, W. Dong, Y. Liu, and J. Chen, "Breaking through the haze: An advanced non-homogeneous dehazing method based on fast Fourier convolution and ConvNeXt," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 1895–1904.

[26] H. Chen et al., "Pre-trained image processing transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 12294–12305.

[27] J. Liang, J. Cao, G. Sun, K. Zhang, L. Van Gool, and R. Timofte, "SwinIR: Image restoration using Swin transformer," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 1833–1844.

[28] B. Benjdira, A. M. Ali, and A. Koubaa, "Streamlined global and local features combinator (SGLC) for high resolution image dehazing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 1855–1864.

[29] S. W. Zamir, A. Arora, S. Khan, M. Hayat, F. S. Khan, and M.-H. Yang, "Restormer: Efficient transformer for high-resolution image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5728–5739.

[30] Z. Wang, X. Cun, J. Bao, W. Zhou, J. Liu, and H. Li, "Uformer: A general U-shaped transformer for image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 17662–17672.

[31] H. Sim et al., "High-resolution image dehazing with respect to training losses and receptive field sizes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 912–919.

[32] S. Ki et al., "Fully end-to-end learning based conditional boundary equilibrium GAN with receptive field sizes enlarged for single ultra-high resolution image dehazing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 817–824.

[33] S. Bianco, L. Celona, F. Piccoli, and R. Schettini, "High-resolution single image dehazing using encoder–decoder architecture," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2019, pp. 1927–1935.

[34] X. Chen, Y. Li, L. Dai, and C. Kong, "Hybrid high-resolution learning for single remote sensing satellite image dehazing," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[35] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10674–10685.

[36] Z. Luo, F. K. Gustafsson, Z. Zhao, J. Sjölund, and T. B. Schön, "Refusion: Enabling large-size realistic image restoration with latent-space diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 1680–1691.

[37] X. Qin, Z. Wang, Y. Bai, X. Xie, and H. Xie, "FFA-Net: Feature fusion attention network for single image dehazing," in *Proc. AAAI Conf. Artif. Intell.*, Feb. 2020, pp. 11908–11915.

[38] Y. Guo et al., "SCANet: Self-paced semi-curricular attention network for non-homogeneous image dehazing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 1885–1894.

[39] W. Dong et al., "End-to-end detail-enhanced dehazing network for remote sensing images," *Remote Sens.*, vol. 16, no. 2, p. 225, Jan. 2024.

[40] B. Huang, Z. Li, C. Yang, F. Sun, and Y. Song, "Single satellite optical imagery dehazing using SAR image prior based on conditional generative adversarial networks," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2020, pp. 1795–1802.

[41] H. Li, Y. Zhang, J. Liu, and Y. Ma, "GTMNet: A vision transformer with guided transmission map for single remote sensing image dehazing," *Sci. Rep.*, vol. 13, no. 1, p. 9222, Jun. 2023.

[42] Y. Cui, W. Ren, X. Cao, and A. Knoll, "Revitalizing convolutional network for image restoration," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 9423–9438, Dec. 2024.

[43] Z. Lihe, J. He, Q. Yuan, X. Jin, Y. Xiao, and L. Zhang, "PhDnet: A novel physic-aware dehazing network for remote sensing images," *Inf. Fusion*, vol. 106, Jun. 2024, Art. no. 102277.

[44] M. Yu, S. Xu, H. Sun, Y. Zheng, and W. Yang, "Hierarchical slice interaction and multi-layer cooperative decoding networks for remote sensing image dehazing," *Image Vis. Comput.*, vol. 148, Aug. 2024, Art. no. 105129.

[45] Y. Huang, Z. Lin, S. Xiong, and T. Sun, "Diffusion models based null-space learning for remote sensing image dehazing," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[46] Z. Luo, F. K. Gustafsson, Z. Zhao, J. Sjölund, and T. B. Schön, "Image restoration with mean-reverting stochastic differential equations," 2023, *arXiv:2301.11699*.

[47] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, "Score-based generative modeling through stochastic differential equations," 2020, *arXiv:2011.13456*.

[48] L. Chen, X. Chu, X. Zhang, and J. Sun, "Simple baselines for image restoration," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 17–33.

[49] P. J. Burt and E. H. Adelson, "The Laplacian pyramid as a compact image code," in *Readings in Computer Vision*. Amsterdam, The Netherlands: Elsevier, 1987, pp. 671–679.

[50] S. Paris, S. W. Hasinoff, and J. Kautz, "Local Laplacian filters: Edge-aware image processing with a Laplacian pyramid," *ACM Trans. Graph.*, vol. 30, no. 4, p. 68, 2011.

[51] C. Ma, Y. Rao, J. Lu, and J. Zhou, "Structure-preserving image super-resolution," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7898–7911, Nov. 2022.

[52] B. Li, X. Peng, Z. Wang, J. Xu, and D. Feng, "AOD-Net: All-in-one dehazing network," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4780–4788.

[53] Y. Li and X. Chen, "A coarse-to-fine two-stage attentive network for haze removal of remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 10, pp. 1751–1755, Oct. 2021.

[54] K. Mei, A. Jiang, J. Li, and M. Wang, "Progressive feature fusion network for realistic image dehazing," in *Proc. Asian Conf. Comput. Vis.*, 2018, pp. 203–215.

[55] X. Liu, Y. Ma, Z. Shi, and J. Chen, "GridDehazeNet: Attention-based multi-scale network for image dehazing," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 7314–7323.

[56] A. Kulkarni and S. Murala, "Aerial image dehazing with attentive deformable transformers," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 6305–6314.

[57] H. Dong, T. Song, X. Qi, G. Jin, J. Jin, and L. Ma, "Prompt-guided sparse transformer for remote sensing image dehazing," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[58] A. Kulkarni, S. S. Phutke, and S. Murala, "Unified transformer network for multi-weather image restoration," in *Proc. Eur. Conf. Comput. Vis.*, 2023, pp. 344–360.

[59] Y. Cui, W. Ren, X. Cao, and A. Knoll, "Focal network for image restoration," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 12955–12965.

[60] H. Sun et al., "Partial Siamese with multiscale bi-codec networks for remote sensing image haze removal," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4106516, doi: 10.1109/TGRS.2023.3321307.

[61] Y. Huang and S. Xiong, "Remote sensing image dehazing using adaptive region-based diffusion models," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[62] L. Yang, J. Cao, H. Wang, S. Dong, and H. Ning, "Hierarchical semantic-guided contextual structure-aware network for spectral satellite image dehazing," *Remote Sens.*, vol. 16, no. 9, p. 1525, Apr. 2024.

[63] H. Zhou, X. Wu, H. Chen, X. Chen, and X. He, "RSDehamba: Lightweight vision mamba for remote sensing satellite image dehazing," 2024, *arXiv:2405.10030*.



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Prof. Koubaa is a Senior Fellow of the Higher Education Academy (U.K.). His prestigious accolades include the Shoman Award for Arab Researchers in 2024 for excellence in AI and robotics, the Third Prize in the SDAIA ALLAM Competition in 2024, selected from over 170 international teams for his contributions to AI innovation, and numerous other awards, such as the AI Leadership Award in 2022 and the KAUST Challenge Award in 2021. He is an Editor of over 20 Springer publications, his exceptional work has earned him global recognition. He has been recognized among the career-based top 2% scientists by Stanford University.

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